

# SoC DESIGN – WEEK 5

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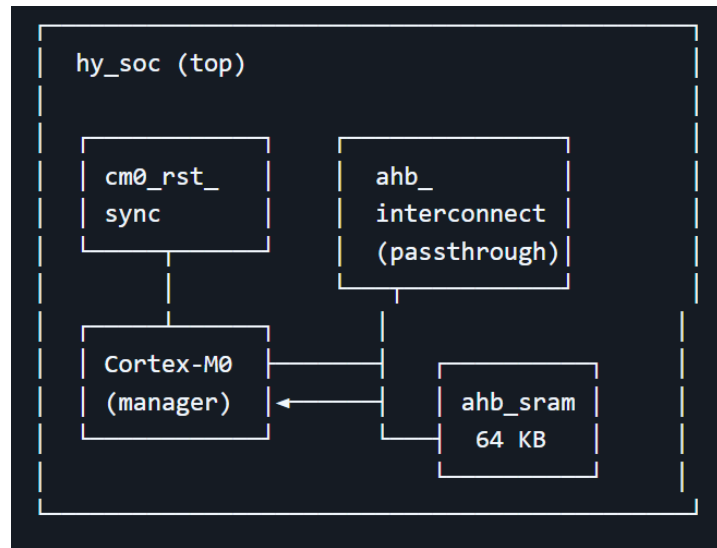
Professor: Ji-Hoon Kim

# Lab0

- Cortex-M0 + SRAM
  - A minimal SoC consisting of ARM Cortex-M0 processor and 64KB SRAM.
  - This is the first step toward understanding AHB-Lite bus operation.

## Memory Map

| Address Range             | Device      | Notes                                          |
|---------------------------|-------------|------------------------------------------------|
| 0x0000_0000 – 0xFFFF_FFFF | SRAM (64KB) | Entire address space maps to SRAM (no decoder) |



# Directory Structure

```
hy_soc/
├── edu/                                Educational labs (AHB-Lite, FPGA target)
│   ├── lab0_m0_sram/                  CM0 + SRAM (passthrough interconnect)
│   ├── lab1_m0_default_slv/          + AHB Default Slave
│   ├── lab2_m0_led/                  + LED Controller (SysTick interrupt)
│   ├── lab3_m0_uart/                  + CMSDK APB UART
│   └── lab4_m0_boot/                  + Boot ROM + SYSCCTRL remap + bootloader
├── common/                            Reusable IP modules
│   ├── ahb_sram/                     AHB-Lite SRAM controller
│   ├── ahb_bus/                     AHB default slave
│   ├── brom/                        AHB Boot ROM + hex generator
│   ├── uart/                        AHB UART (CMSDK APB UART wrapper)
│   ├── led/                         AHB LED controller
│   ├── sysctrl/                     AHB SYSCCTRL (remap + SOC_ID)
│   ├── rst/                         CM0 reset synchronizer
│   ├── cmsis/                       CMSIS Core headers (core_cm0.h)
│   ├── include/                     hy_soc.h (CMSIS-style device header)
│   ├── sim_models/                  uart_mon.v (simulation only)
│   └── arm_ip/                      ARM DesignStart placeholder (see below)
```

```
lab0_m0_sram/
├── rtl/
│   ├── hy_soc.v                     ← Top-level module
│   └── ahb_interconnect.v           ← AHB passthrough (direct to SRAM)
├── tb/
│   └── tb_hy_soc.v                  ← Testbench (PASS/FAIL detection)
├── sw/
│   ├── test.c                      ← Test firmware
│   ├── gcc/                        ← GCC build (startup.s, link.ld, Makefile)
│   └── arm/                         ← ARM Compiler build (scatter.sct, startup_cm0.S)
└── Makefile                        ← Unified build/simulation Makefile
```

# ARM DesignStart

- common/arm\_ip
  - <https://developer.arm.com/downloads/search?entitled=true>

The screenshot displays the ARM DesignStart search interface. On the left, a sidebar contains a filter section with a checkbox labeled "Only show products I have access to" (highlighted with a red box) and several expandable categories: Programme, Product Line, Application, Taxonomy, and Product Group. The main search area has a search bar containing "Cortex-M0" and a search icon. Below the search bar, the text "Search Results" is followed by "Results 1-20 of 57 for Cortex-M0". The results are sorted by "Relevance" and "A-Z". The first result is "Cortex-M0+ Processor (AT590)" with the description "ARMv6-M / Processor Architecture / 3E991" and a "Product Enquiry" button. The second result is "Cortex-M0 DesignStart Evaluation (AT511)" with the description "DevSys ATA Succ Kits / Infrastructure and IO / 3E991" (highlighted with a red box). On the right, a detailed view for "AT511 - Cortex-M0 DesignStart Eval" is shown, including the Name, Filename, and MD5, a file size of 4.1 MB, an "Add to queue" button, and a "Download Now" button (highlighted with a red box).

# ARM DesignStart

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- common/arm\_ip/**cm0\_ds**
  - <https://developer.arm.com/downloads/search?entitled=true>
  - \AT511-r2p0-00rel0-1\AT510-MN-80001-r2p0-00rel0\cores\cortexm0\_designstart\_r2p0\logical\cortexm0\_integration\verilog
    - cortexm0ds\_logic.v
    - CORTEXM0INTEGRATION.v
  - Copy those codes in common/arm\_ip/**cm0\_ds**

# ARM DesignStart

- common/arm\_ip
  - scp .\Cortex-M0 DesignStart.gz {name}@166.104.35.66:~/
  - CORTEX0INTEGRATION.v, cortexm0ds\_logic.v를 common/arm\_ip 경로로 이동

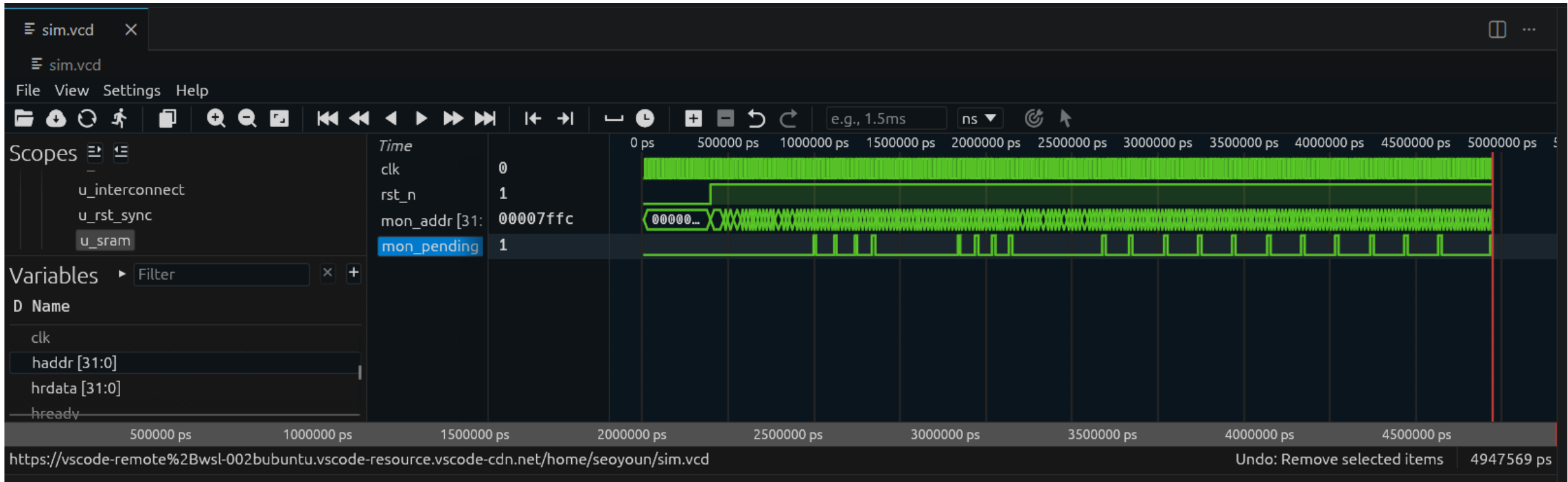
```
[seoyoun@prometheus verilog]$ pwd
/data/home/seoyoun/ca_soc/AT510-MN-80001-r2p0-00rel0/cores/cortexm0_designstart_r2p0/logical/cortexm0_integration/verilog
[seoyoun@prometheus verilog]$ ls
CORTEXM0INTEGRATION.v cortexm0ds_logic.v
```

- ahb\_sram

```
initial begin
    //$readmemh("code.hex", mem);
    $readmemh("C:/Users/dsald/Desktop/hy-soc/edu/lab0_m0_sram/sw/gcc/code.hex", mem);
end
```

# Lab0

- Waveform



# To-Do

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- Simulation with the pre-compiled code
- Simulation with the code compiled with MDK Keil
- FPGA Board Demo



# To-Do

```
module hy_soc (  
    input wire clk,        // system clock  
    input wire rst_n,      // async assert, sync deassert (active-low)  
    output wire [3:0] led  
);
```

```
localparam DONE_ADDR = 32'h00007FFC;  
localparam PASS_CODE = 32'h900DD00D;  
localparam FAIL_CODE = 32'hDEADDEAD;
```

```
reg        mon_pending;  
reg [31:0] mon_addr;
```



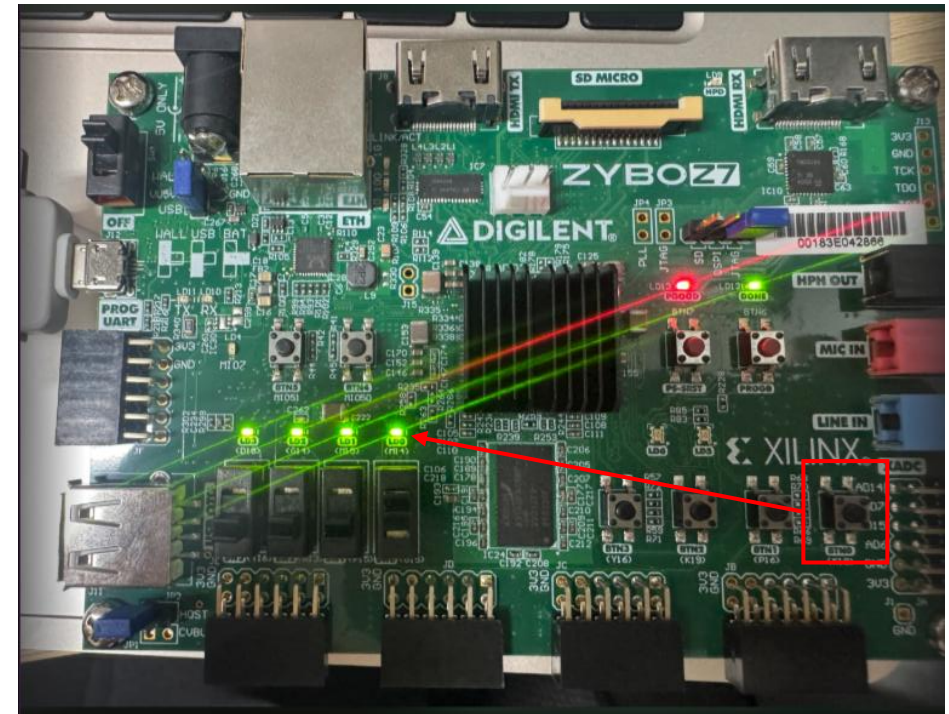
```
// state  
reg [3:0] c_led, n_led;
```



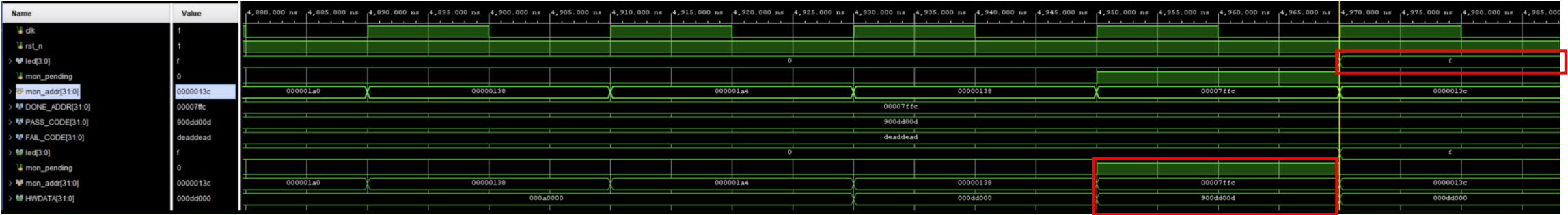
```
assign led = c_led;
```

Turn LEDs on when  
PASS\_CODE(32'h900DD00D) is  
written in  
DONE\_ADDR(32'h00007FFC)

Hint (refer to the testbench)



# To-Do



# THANK YOU

Any questions or comments:  [Issu07@hanyang.ac.kr](mailto:Issu07@hanyang.ac.kr)